Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (`NTL-LTER_Lake_ChemistryPhysics_Raw.csv`). Set date columns to date objects.

2. Build a ggplot theme and set it as your default theme.

```
#1
install.packages("tidyverse")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

```
install.packages("agricolae")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(agricolae)
library(ggplot2)

getwd()
```

```
## [1] "/home/guest/GITClone/EDE_Fall2025"
```

```r
Lake_ChemPhys_Raw <-
  read.csv("/home/guest/GITClone/EDE_Fall2025/Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv",
           stringsAsFactors = TRUE)

Lake_ChemPhys_Raw$sampledate <-
  as.Date(Lake_ChemPhys_Raw$sampledate,
          format = "%m/%d/%y")

head(Lake_ChemPhys_Raw$sampledate)
```

```
## [1] "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27"
## [6] "1984-05-27"
```

```r
#2
my_theme <-
  theme(
    plot.background = element_rect(fill = "white"),
    legend.background = element_rect(fill = "white"),
    legend.title = element_text(color = "turquoise4",
                                face = "bold"),
    legend.text = element_text(color = "turquoise4"),
    panel.background = element_rect(fill = "white"),
    panel.grid.major = element_line(color = "turquoise4"),
    panel.grid.minor = element_line(color = "turquoise4"),
    plot.title = element_text(face = "bold",
                              color = "turquoise4"),
    axis.title.x  = element_text(
                               face = "italic",
                               size = 14,
                               color = "turquoise4"),
    axis.title.y = element_text(
                               face = "italic",
                               size = 14,
                               color = "turquoise4"),
  )
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: Mean lake temperatures recorded during July does not change with depth across all lakes. Ha: Mean lake temperatures recorded during July change with depth across all lakes.

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:

- Only dates in July.
- Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
- Only complete cases (i.e., remove NAs)

5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```r
#4
Lake_ChemPhys <-
  Lake_ChemPhys_Raw%>%
  subset(month(sampledate)==7)%>%
  select(lakename,
         year4,
         daynum,
         depth,
         temperature_C)%>%
  drop_na()

#5
library(viridis)
```

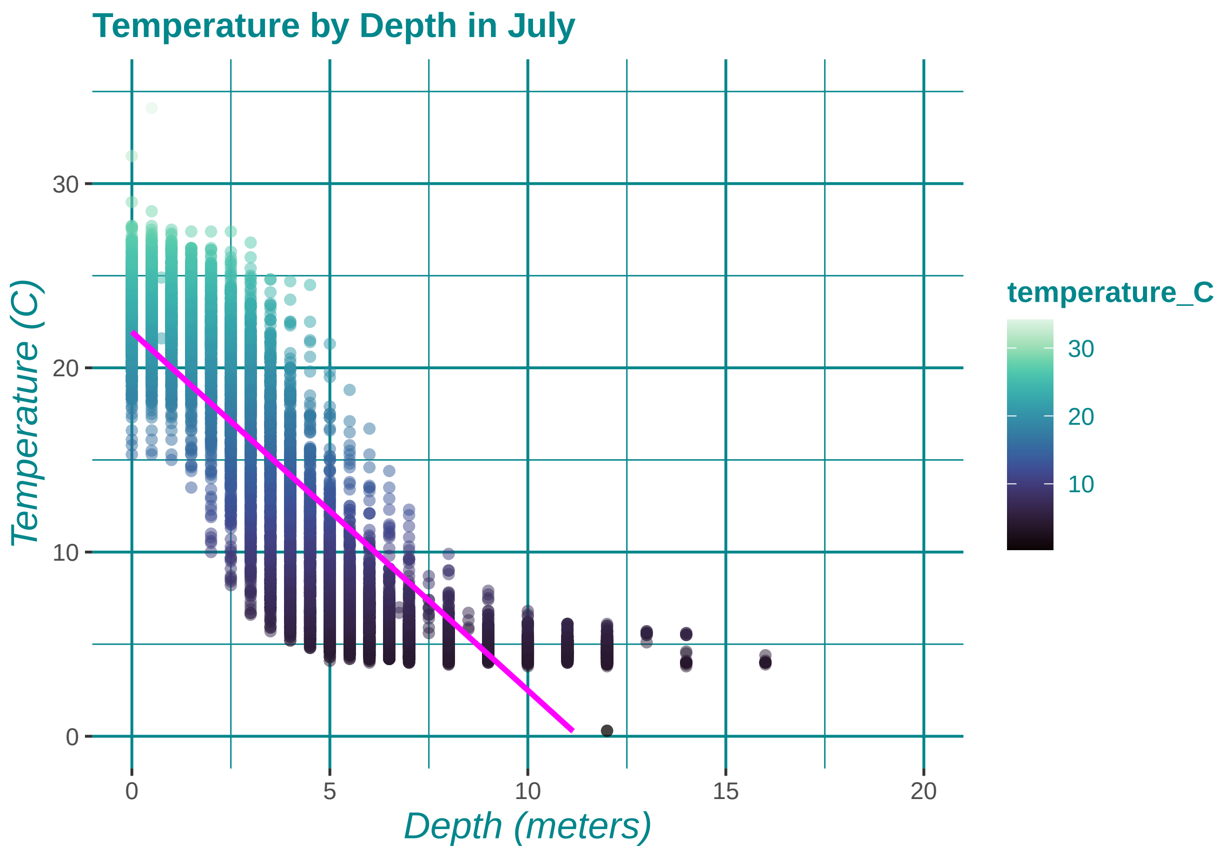
```
## Loading required package: viridisLite
```

```r
Lake_ChemPhys_Plot <-
  ggplot(Lake_ChemPhys,
         aes(x = depth,
             y = temperature_C,
             color = temperature_C)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm",
              se = TRUE,
              color = "magenta") +
  xlim(0,20)+
  ylim(0,35)+
  labs(
    title = "Temperature by Depth in July",
    x = "Depth (meters)",
    y = "Temperature (C)")+
   scale_color_viridis(option = "mako", direction = 1) +
  my_theme

print(Lake_ChemPhys_Plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 24 rows containing missing values or values outside the scale range
## (`geom_smooth()`).
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

   Answer: The figure suggests that in July, as depth increases, temperature decreasees within the lake. The lowest temperatures plateau at around 4 degrees C. So, at a certain point, depth doesn't have any increased impact on the temperature of the water.

7. Perform a linear regression to test the relationship and display the results.

```r
#7
tempdepth.regression <- lm(
  data = Lake_ChemPhys,
  temperature_C ~ depth)

summary(tempdepth.regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = Lake_ChemPhys)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3   <2e-16 ***
## depth       -1.94621    0.01174  -165.8   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

   Answer: With an R-squared value of 0.7387, about 73.87% of the variability in temperature is explained by changes in depth. This finding is based on 9726 degrees of freedom which means that there was a large, sufficient sample size to observe this relationship. These results are also statistically significant with a p-value of <2.2e-16, which is far within the threshold of statistical significance: 0.05. With every 1m increase in depth, temperature decreases by around 1.94621 degrees C.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.

10. Run a multiple regression on the recommended set of variables.

```
#9
library(dplyr)
#install.packages("corrplot")
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
Lake_ChemPhys_sub <-
 Lake_ChemPhys %>%
  select(temperature_C:year4) %>%
```

```r
 na.omit()

Lake_ChemPhys_AIC<- lm(data = Lake_ChemPhys_sub,
temperature_C ~ depth + daynum + year4)

step(Lake_ChemPhys_AIC)
```

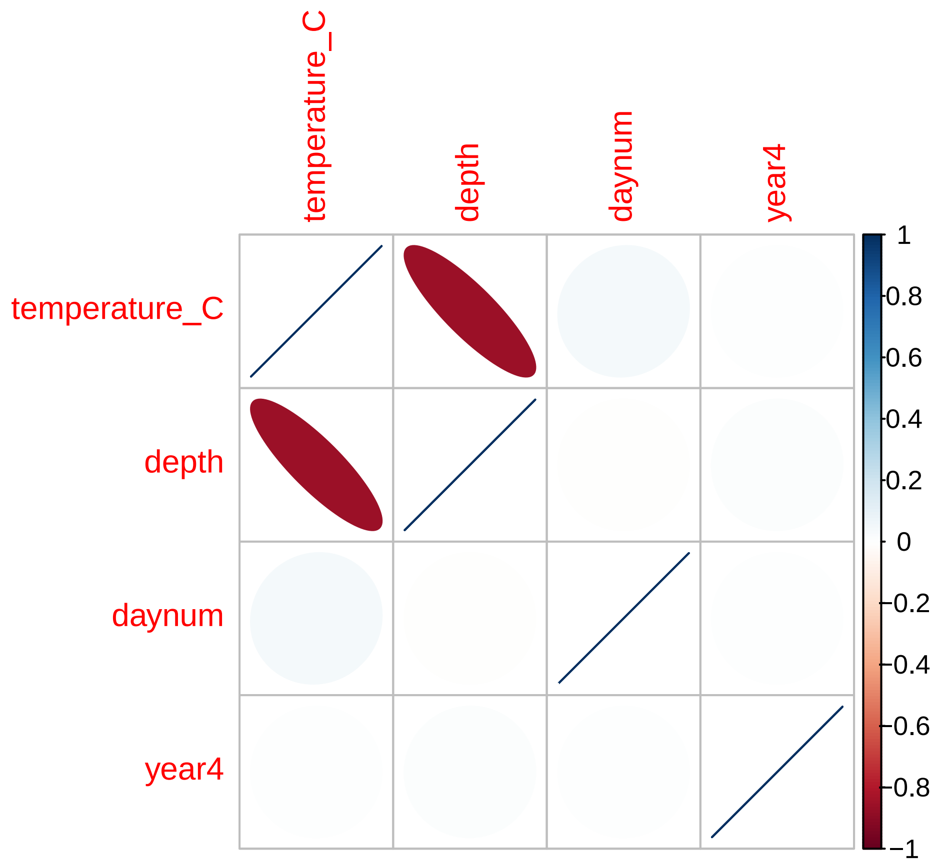
```
## Start:  AIC=26065.53
## temperature_C ~ depth + daynum + year4
##
##          Df Sum of Sq    RSS   AIC
## <none>                141687 26066
## - year4   1       101 141788 26070
## - daynum  1      1237 142924 26148
## - depth   1    404475 546161 39189
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + daynum + year4, data = Lake_ChemPhys_sub)
##
## Coefficients:
## (Intercept)        depth       daynum        year4
##    -8.57556     -1.94644      0.03978      0.01134
```

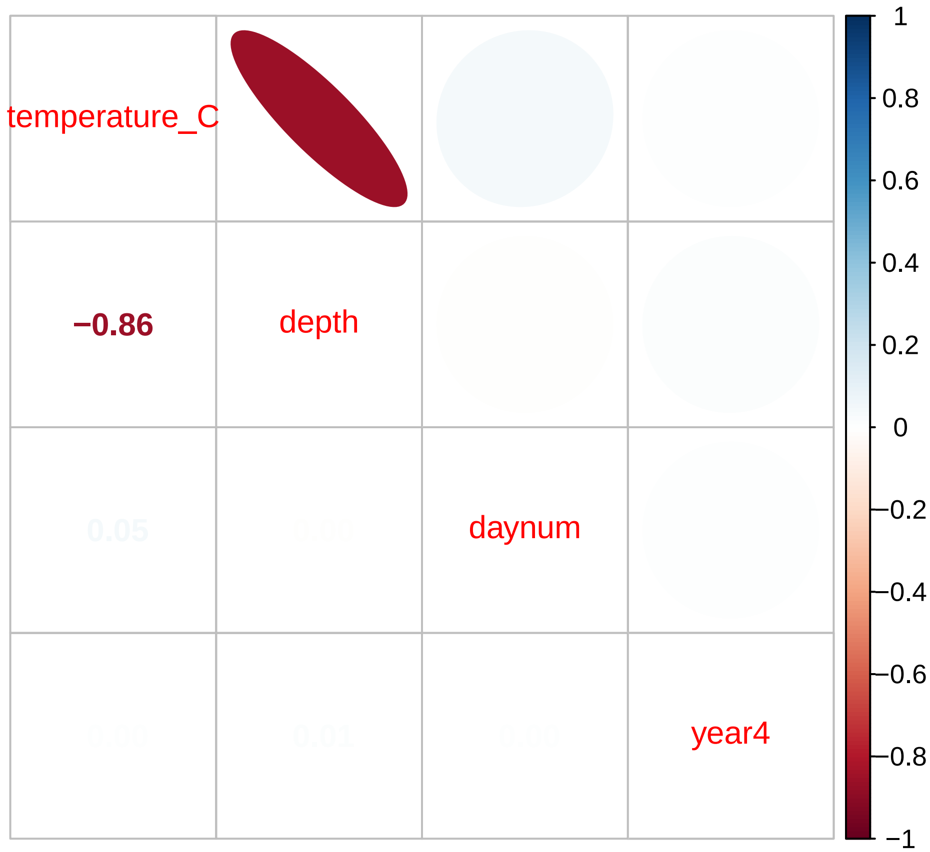
```r
#Corrplot for fun

Lake_ChemPhys_Corr <- cor(Lake_ChemPhys_sub)

corrplot(Lake_ChemPhys_Corr,
         method = "ellipse")
```




```
corrplot.mixed(Lake_ChemPhys_Corr,
               upper = "ellipse")
```




```r
#10

Lake_ChemPhys_BestRegression<- lm(data = Lake_ChemPhys_sub,
temperature_C ~ depth + daynum + year4)

summary(Lake_ChemPhys_BestRegression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + daynum + year4, data = Lake_ChemPhys_sub)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715   -0.994  0.32044
## depth       -1.946437   0.011683 -166.611  < 2e-16 ***
## daynum       0.039780   0.004317    9.215  < 2e-16 ***
## year4        0.011345   0.004299    2.639  0.00833 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
```

```
## F-statistic:  9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

    Answer: The final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression are depth, daynum, and year4. This model explains 74.1% of the observed variance with an R-squared value of 0.741. This is a slight improvement over the model using only depth, as that previous model could explain ~0.23% less of the observed variance.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```r
#12
Lake_ChemPhys_ANOVA <- aov(data = Lake_ChemPhys,
                           temperature_C ~ lakename)
summary(Lake_ChemPhys_ANOVA)
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## lakename       8  21642  2705.2      50 <2e-16 ***
## Residuals   9719 525813    54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
Lake_ChemPhys_LM <- lm(data = Lake_ChemPhys,
                       temperature_C ~ lakename)
summary(Lake_ChemPhys_LM)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = Lake_ChemPhys)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               17.6664     0.6501  27.174  < 2e-16 ***
## lakenameCrampton Lake     -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake    -7.3987     0.6918 -10.695  < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake         -3.8522     0.6656  -5.788 7.36e-09 ***
```

```
## lakenamePeter Lake        -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake      -6.5972     0.6769  -9.746  < 2e-16 ***
## lakenameWard Lake         -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake    -6.0878     0.6895  -8.829  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

    Answer: Yes, there is a significant difference in mean temperature among the lakes. From both the linear model and ANOVA test, the adjusted p-value for the difference in mean temperature among the lakes is statistically significant at <2e-16 (much less than 0.05 or 0.001).

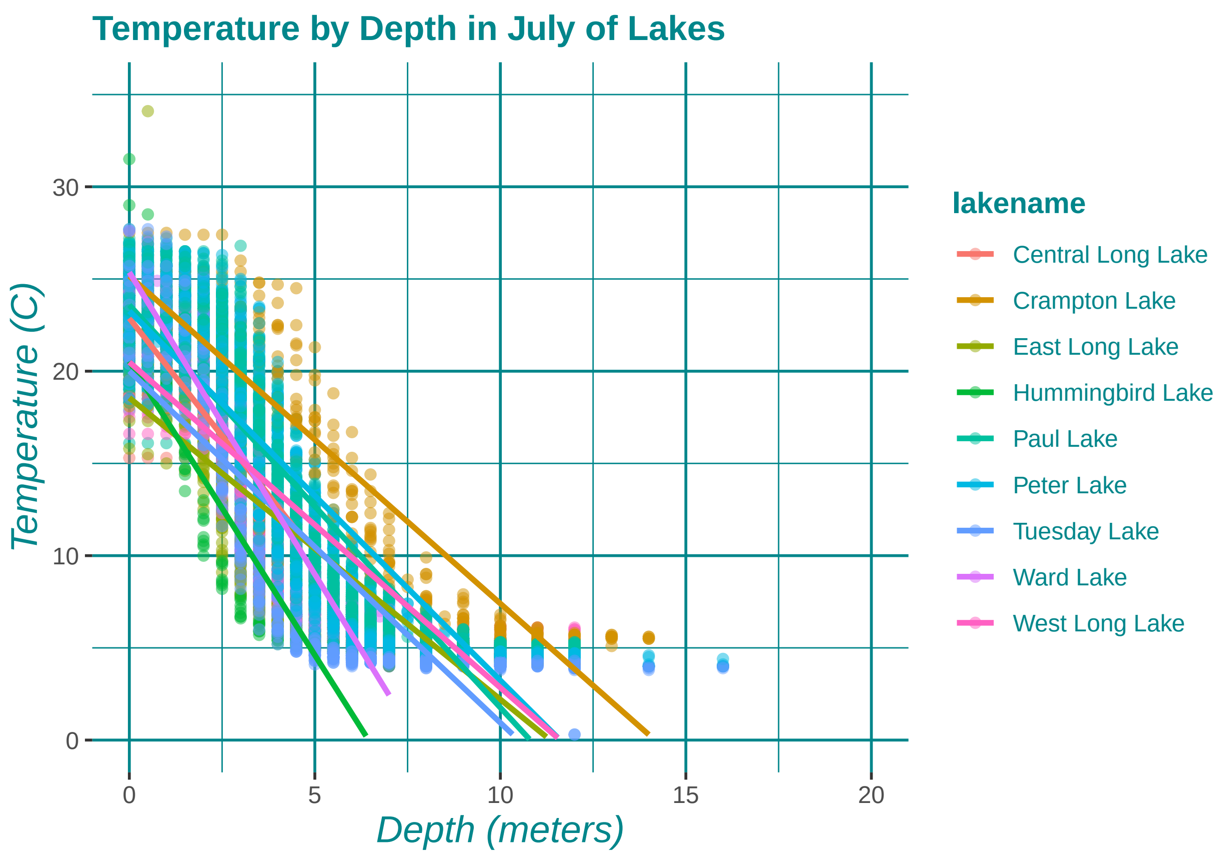
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a geom_smooth (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```r
#14.
library(ggplot2)
Lake_ChemPhys_Plot2 <-
  ggplot(Lake_ChemPhys,
         aes(x = depth,
             y = temperature_C,
             color = lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm",
              se = FALSE,
              aes(color = lakename)) +
  xlim(0,20)+
  ylim(0,35)+
  labs(
    title = "Temperature by Depth in July of Lakes",
    x = "Depth (meters)",
    y = "Temperature (C)")+
  my_theme

print(Lake_ChemPhys_Plot2)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## (`geom_smooth()`).
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
TukeyHSD(Lake_ChemPhys_ANOVA)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = Lake_ChemPhys)
##
## $lakename
##                                        diff        lwr        upr     p adj
## Crampton Lake-Central Long Lake    -2.3145195 -4.7031913  0.0741524 0.0661566
## East Long Lake-Central Long Lake   -7.3987410 -9.5449411 -5.2525408 0.0000000
## Hummingbird Lake-Central Long Lake -6.8931304 -9.8184178 -3.9678430 0.0000000
## Paul Lake-Central Long Lake        -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake       -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake     -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake        -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake   -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake       -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake     -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake            -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake           -2.0356263 -3.3842699 -0.6869828 0.0000999
## Tuesday Lake-Crampton Lake         -4.2826611 -5.6895065 -2.8758157 0.0000000
```

```
## Ward Lake-Crampton Lake           -0.8932661 -3.3684639  1.5819317 0.9714459
## West Long Lake-Crampton Lake      -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake    0.5056106 -1.7364925  2.7477137 0.9988050
## Paul Lake-East Long Lake           3.5465903  2.6900206  4.4031601 0.0000000
## Peter Lake-East Long Lake          3.0485952  2.2005025  3.8966879 0.0000000
## Tuesday Lake-East Long Lake        0.8015604 -0.1363286  1.7394495 0.1657485
## Ward Lake-East Long Lake           4.1909554  1.9488523  6.4330585 0.0000002
## West Long Lake-East Long Lake      1.3109897  0.2885003  2.3334791 0.0022805
## Paul Lake-Hummingbird Lake         3.0409798  0.8765299  5.2054296 0.0004495
## Peter Lake-Hummingbird Lake        2.5429846  0.3818755  4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake      0.2959499 -1.9019508  2.4938505 0.9999752
## Ward Lake-Hummingbird Lake         3.6853448  0.6889874  6.6817022 0.0043297
## West Long Lake-Hummingbird Lake    0.8053791 -1.4299320  3.0406903 0.9717297
## Peter Lake-Paul Lake              -0.4979952 -1.1120620  0.1160717 0.2241586
## Tuesday Lake-Paul Lake            -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake                0.6443651 -1.5200848  2.8088149 0.9916978
## West Long Lake-Paul Lake          -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake           -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake               1.1423602 -1.0187489  3.3034693 0.7827037
## West Long Lake-Peter Lake         -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake             3.3893950  1.1914943  5.5872956 0.0000609
## West Long Lake-Tuesday Lake        0.5094292 -0.4121051  1.4309636 0.7374387
## West Long Lake-Ward Lake          -2.8799657 -5.1152769 -0.6446546 0.0021080
```

16.From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

> Answer: Ward Lake and Peter Lake have the same mean temperature, statistically speaking, with a p-value of 0.7827, which is not statistically significant. Paul Lake and Peter Lake also have the same mean temperature with a statistically insignificant p-value of 0.2242. There is no lake that has a mean temperature that is statistically distinct from all other lakes. Each lake has at least one statistically insignificant pairing with another lake (p-value greater than 0.05).

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

> Answer: If we were just looking at Peter and Paul Lake, another test we might explore to see whether they have distinct mean temperatures is a two-sample t-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```r
ChemPhys_Crampton_Ward <-
  Lake_ChemPhys%>%
  filter(lakename == c("Crampton Lake", "Ward Lake"))

ChemPhys_Crampton_Ward
```

```
##          lakename year4 daynum depth temperature_C
## 1   Crampton Lake  1999    196  0.50          22.6
```

```
## 2   Crampton Lake  1999    196  1.50          22.2
## 3   Crampton Lake  1999    196  2.50          21.9
## 4   Crampton Lake  1999    196  3.50          21.3
## 5   Crampton Lake  1999    196  4.50          17.1
## 6   Crampton Lake  1999    196  5.50          12.1
## 7   Crampton Lake  1999    196  6.50           9.1
## 8   Crampton Lake  1999    196  8.00           7.1
## 9   Crampton Lake  1999    196 10.00           5.8
## 10  Crampton Lake  1999    196 12.00           5.4
## 11  Crampton Lake  2003    183  0.50          21.9
## 12  Crampton Lake  2003    183  1.50          21.8
## 13  Crampton Lake  2003    183  2.50          21.3
## 14  Crampton Lake  2003    183  3.50          19.9
## 15  Crampton Lake  2003    183  4.50          15.2
## 16  Crampton Lake  2003    183  5.50          11.2
## 17  Crampton Lake  2003    183  6.50           8.8
## 18  Crampton Lake  2003    183  8.00           6.8
## 19  Crampton Lake  2003    183 10.00           5.4
## 20  Crampton Lake  2003    183 12.00           5.0
## 21  Crampton Lake  2003    197  0.50          21.9
## 22  Crampton Lake  2003    197  1.50          21.8
## 23  Crampton Lake  2003    197  2.50          21.7
## 24  Crampton Lake  2003    197  3.50          21.6
## 25  Crampton Lake  2003    197  4.50          17.3
## 26  Crampton Lake  2003    197  5.50          12.5
## 27  Crampton Lake  2003    197  6.50           9.8
## 28  Crampton Lake  2003    197  8.00           7.0
## 29  Crampton Lake  2003    197 10.00           5.5
## 30  Crampton Lake  2003    197 12.00           5.1
## 31  Crampton Lake  2003    211  0.00          23.2
## 32  Crampton Lake  2003    211  1.00          23.2
## 33  Crampton Lake  2003    211  2.00          23.3
## 34  Crampton Lake  2003    211  3.00          23.3
## 35  Crampton Lake  2003    211  4.00          22.4
## 36  Crampton Lake  2003    211  5.00          17.3
## 37  Crampton Lake  2003    211  6.00          12.1
## 38  Crampton Lake  2003    211  7.00           9.6
## 39  Crampton Lake  2003    211  9.00           6.4
## 40  Crampton Lake  2003    211 11.00           5.4
## 41  Crampton Lake  2004    183  0.00          19.8
## 42  Crampton Lake  2004    183  1.00          19.8
## 43  Crampton Lake  2004    183  2.00          19.7
## 44  Crampton Lake  2004    183  3.00          18.9
## 45  Crampton Lake  2004    183  4.00          18.0
## 46  Crampton Lake  2004    183  5.00          15.6
## 47  Crampton Lake  2004    183  6.00          11.2
## 48  Crampton Lake  2004    183  7.00           9.0
## 49  Crampton Lake  2004    183  9.00           6.6
## 50  Crampton Lake  2004    183 11.00           5.6
## 51  Crampton Lake  2004    183 13.00           5.5
## 52  Crampton Lake  2004    191  0.00          18.1
## 53  Crampton Lake  2004    191  1.00          18.0
## 54  Crampton Lake  2004    191  2.00          17.9
## 55  Crampton Lake  2004    191  3.00          17.9
```

```
## 56  Crampton Lake  2004    191  4.00          17.6
## 57  Crampton Lake  2004    191  5.00          16.6
## 58  Crampton Lake  2004    191  6.00          12.1
## 59  Crampton Lake  2004    191  7.00           9.4
## 60  Crampton Lake  2004    191  9.00           6.6
## 61  Crampton Lake  2004    191 11.00           5.7
## 62  Crampton Lake  2004    191 13.00           5.5
## 63  Crampton Lake  2004    197  0.50          21.8
## 64  Crampton Lake  2004    197  1.50          21.6
## 65  Crampton Lake  2004    197  2.50          20.9
## 66  Crampton Lake  2004    197  3.50          19.1
## 67  Crampton Lake  2004    197  4.50          17.9
## 68  Crampton Lake  2004    197  5.50          14.8
## 69  Crampton Lake  2004    197  6.50          11.1
## 70  Crampton Lake  2004    197  8.00           7.7
## 71  Crampton Lake  2004    197 10.00           6.1
## 72  Crampton Lake  2004    197 12.00           5.6
## 73  Crampton Lake  2004    197 14.00           5.5
## 74  Crampton Lake  2004    204  0.00          23.8
## 75  Crampton Lake  2004    204  1.00          23.8
## 76  Crampton Lake  2004    204  2.00          23.8
## 77  Crampton Lake  2004    204  3.00          22.0
## 78  Crampton Lake  2004    204  4.00          19.9
## 79  Crampton Lake  2004    204  5.00          17.3
## 80  Crampton Lake  2004    204  6.00          13.5
## 81  Crampton Lake  2004    204  7.00          10.1
## 82  Crampton Lake  2004    204  9.00           6.8
## 83  Crampton Lake  2004    204 11.00           5.8
## 84  Crampton Lake  2004    204 13.00           5.6
## 85  Crampton Lake  2004    211  0.00          22.7
## 86  Crampton Lake  2004    211  1.00          22.7
## 87  Crampton Lake  2004    211  2.00          22.7
## 88  Crampton Lake  2004    211  3.00          22.7
## 89  Crampton Lake  2004    211  4.00          20.8
## 90  Crampton Lake  2004    211  5.00          17.5
## 91  Crampton Lake  2004    211  6.00          13.6
## 92  Crampton Lake  2004    211  7.00          10.3
## 93  Crampton Lake  2004    211  9.00           6.8
## 94  Crampton Lake  2004    211 11.00           5.8
## 95  Crampton Lake  2004    211 13.00           5.6
## 96  Crampton Lake  2005    186  0.00          22.3
## 97  Crampton Lake  2005    186  1.00          22.4
## 98  Crampton Lake  2005    186  2.00          22.4
## 99  Crampton Lake  2005    186  3.00          22.4
## 100 Crampton Lake  2005    186  4.00          22.3
## 101 Crampton Lake  2005    186  5.00          17.9
## 102 Crampton Lake  2005    186  6.00          13.5
## 103 Crampton Lake  2005    186  7.00          10.8
## 104 Crampton Lake  2005    186  9.00           7.4
## 105 Crampton Lake  2005    186 11.00           5.8
## 106 Crampton Lake  2005    192  0.00          25.7
## 107 Crampton Lake  2005    192  1.00          25.4
## 108 Crampton Lake  2005    192  2.00          25.3
## 109 Crampton Lake  2005    192  3.00          23.5
```

```
## 110 Crampton Lake  2005    192  4.00          22.5
## 111 Crampton Lake  2005    192  5.00          19.5
## 112 Crampton Lake  2005    192  6.00          14.6
## 113 Crampton Lake  2005    192  7.00          11.4
## 114 Crampton Lake  2005    192  9.00           7.5
## 115 Crampton Lake  2005    192 11.00           6.0
## 116 Crampton Lake  2005    192 13.00           5.6
## 117 Crampton Lake  2005    199  0.50          27.5
## 118 Crampton Lake  2005    199  1.50          27.4
## 119 Crampton Lake  2005    199  2.50          27.4
## 120 Crampton Lake  2005    199  3.50          24.8
## 121 Crampton Lake  2005    199  4.50          22.5
## 122 Crampton Lake  2005    199  5.50          17.1
## 123 Crampton Lake  2005    199  6.50          13.5
## 124 Crampton Lake  2005    199  8.00           9.0
## 125 Crampton Lake  2005    199 10.00           6.6
## 126 Crampton Lake  2005    199 12.00           5.7
## 127 Crampton Lake  2005    206  0.00          25.3
## 128 Crampton Lake  2005    206  1.00          25.0
## 129 Crampton Lake  2005    206  2.00          24.9
## 130 Crampton Lake  2005    206  3.00          24.8
## 131 Crampton Lake  2005    206  4.00          24.7
## 132 Crampton Lake  2005    206  5.00          21.3
## 133 Crampton Lake  2005    206  6.00          16.7
## 134 Crampton Lake  2005    206  7.00          12.3
## 135 Crampton Lake  2005    206  9.00           7.9
## 136 Crampton Lake  2005    206 11.00           6.1
## 137 Crampton Lake  2005    206 13.00           5.7
## 138 Crampton Lake  2006    186  0.50          22.8
## 139 Crampton Lake  2006    186  1.50          22.8
## 140 Crampton Lake  2006    186  2.50          22.5
## 141 Crampton Lake  2006    186  3.50          22.6
## 142 Crampton Lake  2006    186  4.50          17.4
## 143 Crampton Lake  2006    186  5.50          13.7
## 144 Crampton Lake  2006    186  6.50          10.8
## 145 Crampton Lake  2006    186  7.50           8.3
## 146 Crampton Lake  2006    186  8.50           6.3
## 147 Crampton Lake  2006    186 10.00           5.5
## 148 Crampton Lake  2006    186 12.00           5.1
## 149 Crampton Lake  2006    200  0.50          25.4
## 150 Crampton Lake  2006    200  1.50          25.4
## 151 Crampton Lake  2006    200  2.50          25.4
## 152 Crampton Lake  2006    200  3.50          24.1
## 153 Crampton Lake  2006    200  4.50          19.8
## 154 Crampton Lake  2006    200  5.50          15.0
## 155 Crampton Lake  2006    200  6.50          11.4
## 156 Crampton Lake  2006    200  7.50           8.7
## 157 Crampton Lake  2006    200  8.50           6.7
## 158 Crampton Lake  2006    200 10.00           5.7
## 159 Crampton Lake  2006    200 12.00           5.1
## 160     Ward Lake  2010    188  0.50          25.3
## 161     Ward Lake  2010    188  1.00          24.8
## 162     Ward Lake  2010    188  2.00          21.8
## 163     Ward Lake  2010    188  3.00          16.6
```

```
## 164     Ward Lake  2010    188  4.00           12.5
## 165     Ward Lake  2010    188  5.00            9.0
## 166     Ward Lake  2010    188  6.00            7.1
## 167     Ward Lake  2010    188  6.75            6.7
## 168     Ward Lake  2010    195  0.50           24.1
## 169     Ward Lake  2010    195  1.50           23.9
## 170     Ward Lake  2010    195  2.50           20.3
## 171     Ward Lake  2010    195  3.50           14.7
## 172     Ward Lake  2010    195  4.50           10.6
## 173     Ward Lake  2010    195  5.50            7.7
## 174     Ward Lake  2010    195  6.50            7.0
## 175     Ward Lake  2010    202  0.50           23.8
## 176     Ward Lake  2010    202  1.50           23.3
## 177     Ward Lake  2010    202  2.50           20.7
## 178     Ward Lake  2010    202  3.50           15.1
## 179     Ward Lake  2010    202  4.50           10.7
## 180     Ward Lake  2010    202  5.50            7.9
## 181     Ward Lake  2010    202  6.50            7.1
## 182     Ward Lake  2010    209  0.00           23.6
## 183     Ward Lake  2010    209  1.00           23.5
## 184     Ward Lake  2010    209  2.00           23.1
## 185     Ward Lake  2010    209  3.00           18.2
## 186     Ward Lake  2010    209  4.00           13.4
## 187     Ward Lake  2010    209  5.00            9.2
## 188     Ward Lake  2010    209  6.00            7.4
## 189     Ward Lake  2010    209  6.75            7.0
## 190     Ward Lake  2012    187  0.50           27.0
## 191     Ward Lake  2012    187  1.50           20.4
## 192     Ward Lake  2012    187  2.50           12.3
## 193     Ward Lake  2012    187  3.50            7.9
## 194     Ward Lake  2012    187  4.50            6.4
## 195     Ward Lake  2012    187  5.50            5.8
## 196     Ward Lake  2012    194  0.50           25.1
## 197     Ward Lake  2012    194  1.50           20.8
## 198     Ward Lake  2012    194  2.50           12.6
## 199     Ward Lake  2012    194  3.50            8.2
## 200     Ward Lake  2012    194  4.50            6.6
## 201     Ward Lake  2012    194  5.50            6.0
## 202     Ward Lake  2012    201  0.50           23.3
## 203     Ward Lake  2012    201  1.50           21.5
## 204     Ward Lake  2012    201  2.50           13.2
## 205     Ward Lake  2012    201  3.50            8.6
## 206     Ward Lake  2012    201  4.50            6.8
## 207     Ward Lake  2012    201  5.50            6.2
## 208     Ward Lake  2012    201  6.50            5.9
## 209     Ward Lake  2012    208  0.50           24.2
## 210     Ward Lake  2012    208  1.50           22.3
## 211     Ward Lake  2012    208  2.50           13.7
## 212     Ward Lake  2012    208  3.50            8.5
## 213     Ward Lake  2012    208  4.50            6.8
## 214     Ward Lake  2012    208  5.50            6.1
## 215     Ward Lake  2012    208  6.50            5.9
```

```r
tail(ChemPhys_Crampton_Ward)
```

```
##      lakename year4 daynum depth temperature_C
## 210 Ward Lake  2012    208   1.5          22.3
## 211 Ward Lake  2012    208   2.5          13.7
## 212 Ward Lake  2012    208   3.5           8.5
## 213 Ward Lake  2012    208   4.5           6.8
## 214 Ward Lake  2012    208   5.5           6.1
## 215 Ward Lake  2012    208   6.5           5.9
```

```r
Crampton_Ward_ttest <- t.test(ChemPhys_Crampton_Ward$temperature_C
                              ~ ChemPhys_Crampton_Ward$lakename)
Crampton_Ward_ttest
```

```
##
##  Welch Two Sample t-test
##
## data:  ChemPhys_Crampton_Ward$temperature_C by ChemPhys_Crampton_Ward$lakename
## t = 0.98673, df = 95.77, p-value = 0.3263
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is n
## 95 percent confidence interval:
##  -1.130614  3.365610
## sample estimates:
## mean in group Crampton Lake     mean in group Ward Lake
##                    15.37107                    14.25357
```

> Answer: This test says that the p-value for this relationship is 0.3263. The mean temperature for Crampton Lake is 15.37 degrees C and the mean temperature for Ward Lake is 14.25 degrees C (unequal). This result does not match my answer for part 16. The post-hoc test found that the p-value between Crampton Lake and Ward Lake was 0.9714459. However, both values are statistically insignificant, so in practice, this doesn't change our acceptance of the null hypothesis in terms of the mean temperature of these two lakes. In other words, the mean July temperature of Crampton and Ward Lake is the same.